

Amyotrophic lateral sclerosis (ALS) Classification using Orange 3

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Abstract—Amyotrophic Lateral Sclerosis (ALS) is a disease that gradually damages nerve cells and weakens motor control, often affecting how a person speaks. Picking up on small changes in the voice early on can be very useful, since it helps doctors reach a diagnosis sooner and improve treatment decisions. In this study, we focus on how people pronounce simple vowels to see whether machine learning can tell ALS patients apart from healthy speakers. The dataset included 128 recordings of the vowels a and i, collected from 64 individuals, with both groups—patients and controls—almost evenly represented. From these samples, we pulled out several acoustic features, such as Mel-Frequency Cepstral Coefficients (MFCC), jitter, shimmer, formant values, and the Harmonics-to-Noise Ratio (HNR). These were then used as input to a set of supervised learning models: Logistic Regression, Support Vector Machines (SVM), Random Forests, Gradient Boosting and Neural Networks. To evaluate the performance of each model, we used markers such as ROC curve analysis, Confusion matrix etc.

Index Terms—Amyotrophic Lateral Sclerosis, vowels, acoustic analysis, supervised learning.

I. INTRODUCTION

ALS is a rare neurological disease that mainly affects motor neurons. Gradually these nerve cells stop working properly, which makes muscles weaker and weaker. In most patients the first sign is trouble with speech, and later swallowing and breathing also become difficult. The average person diagnosed with ALS usually survives only three to five years, which is why being able to notice the illness early is so important.

The way ALS shows up can differ, but there are some common patterns. Many patients develop bulbar issues, meaning speech becomes slurred and swallowing is not smooth. Muscles also waste away, leading to loss of control in the arms or legs. The most dangerous problem is respiratory weakness, which eventually becomes life-threatening. In a smaller group of cases, ALS overlaps with dementia and affects thinking as well.

Researchers have therefore looked for easier alternatives. One promising direction is machine learning, which can analyze simple, non-invasive data such as voice recordings. Even a person holding a vowel sound can carry tiny acoustic signals of muscle decline that the human ear might miss. On evaluating

the acoustic features, we will be able to detect if the subject has ALS or not.

II. METHODOLOGY

A. About the Dataset

The speech dataset was put together at the Republican Research and Clinical Center of Neurology and Neurosurgery in Minsk, Belarus. In total, it contains 128 vowel samples, half of them of the letter "a" and the other half of the letter "i", recorded from 64 people. Out of these, 31 participants had ALS and 33 were healthy, so the database is almost evenly balanced between the two groups. Each person was asked to hold the vowels at a comfortable pitch and loudness for as long as they could manage.

Looking more closely at the ALS group, there were 17 men aged between 40 and 69 years, averaging 61 years, and 14 women aged 39 to 70 years, averaging around 57 years. The healthy controls included 13 men (34–80 years, mean age about 50) and 20 women (37–68 years, average about 56). Recordings were captured on ordinary smartphones with headsets at 44.1 kHz and stored as uncompressed 16-bit PCM audio. On average, healthy controls sustained their vowels for 3.7 seconds, while ALS patients lasted a little longer at about 4.1 seconds. More clinical details on the ALS participants are available in the related study.

B. Feature Extraction

We extracted a set of voice features commonly linked to speech disorder:

MFCCs: extracting the general spectral characteristics of speech.

Jitter: measured pitch stability.

Shimmer: captured fluctuations in loudness.

Formants (F1–F3): demonstrated how the vocal tract shaped the sound.

HNR: computed the clear voice to noise ratio.

GNE: Evaluated the breathiness of the voice.

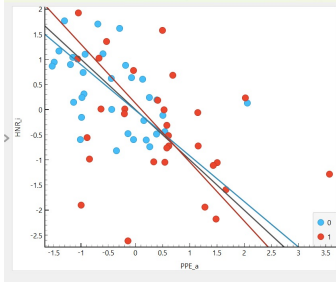


Fig. 1. Scatter Plot

C. Data Visualization

For better insight into data, we visualized the features by scatter plots, box plots, and violin plot in an effort to compare those two groups. Jitter and shimmer were significantly different for ALS individuals, and MFCCs showed significant variations .

III. WORKFLOW DESIGN USING ORANGE 3 SOFTWARE

We started by loading in the data. It was speech recordings, some numerical recording features, and class labels, either ALS or not. Already, something was clearly wrong. There were missing data, some recordings did not complete. We marked those. At the same time, we highlighted early speech impairment for a patient. As one would guess, they occasionally showed very subtle signs. Then we decided which columns were worthwhile to retain. Age and gender did not seem very important at first, so we retained mainly speech-related features like pitch, formants, and clarity scores. The data was cleaned on a few trials. The values were normalized such that one measure did not dominate another. The categorical data was encoded into numbers. Some records did not make the cut and were removed. It was far from perfect, but at least the dataset was tractable. After the feature-cleaning procedure, we looked at which features were significant. Some speech measures changed a great deal for ALS participants, while others did not shift at all. The latter were kept. For modeling, we attempted a few different algorithms. We tried a simple regression first, then tree-based models, a margin-based classifier, and a small neural network. All the models were tested and retested multiple times by splitting the data differently. The tables listing correct versus incorrect predictions were checked multiple times—especially for mildly symptomatic patients that were commonly missed by the models. The true positive rate versus false alarms were also plotted for making decisions on thresholds for early detection of ALS. That seemed to improve sensitivity for subtle speech patterns. Interactive workflow. Choosing points within bar charts or scatter plots spread the selection downstream. This allowed us to probe for patterns, focus feature selection, and find clusters of early “gibberish” speech patients. The end result was noisy recordings and useful predictions. The workflow was far from perfect, yet it captured subtle signals that could be present in early ALS.

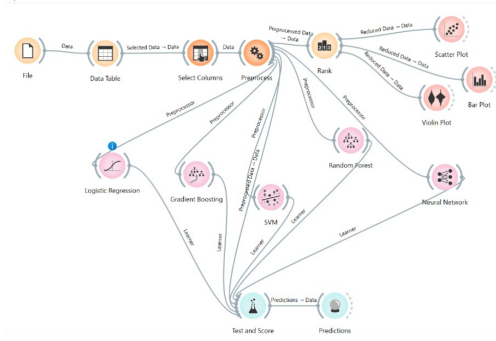


Fig. 2. Orange Workflow

A. Machine Learning Algorithms Used

Our solutions were trained on labeled speech samples, that is, recordings both from people with ALS and healthy people. Our end goal was figuring out which patterns in speech were likely linked to ALS and which looked normal. We did a small amount of data preparation at the beginning. Some data was missing data, and some recordings were partial. We completed those, normalized the readings, and made sure nothing was overpowering the analysis. Then we focused on the meaningful parts—the stuff like pitch, distinctiveness, and how the sounds blended together over time. Scaling kept everything on roughly the same level.

Overview and Working of Algorithms

Logistic Regression We started with a naive probability-based approach that looks at the features and makes predictions about how likely it is that a sample comes from an ALS patient. This was a useful approach because it gave us a sense of what were the most relevant speech features. It was easy to interpret and gave us a nice baseline.

$$P(Y = 1 | X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n)}}$$

Gradient Boosting Then we adopted an idea where a large number of small models are constructed one atop the other, each one attempting to correct errors of the former one. It was useful due to the tendency for speech features to interact in complex fashions. It also accounted for the circumstance that we possessed fewer ALS samples than healthy samples, rendering predictions more trustworthy.

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x)$$

Support Vector Machine This method finds the best boundary between normal speech and ALS. It is extremely helpful if the patterns are weak or non-linear. For example, a mix between clarity and a measure of pitch might exist for ALS, but each one separately might not be distinct.

$$\text{maximize } \frac{2}{\|w\|} \quad \text{subject to } y_i(w \cdot x_i + b) \geq 1, \quad i = 1, \dots, N$$

Random Forest Here, multiple decision trees were built on different subsets of data. Each tree cast a “vote,” and the tree

that got the majority vote was the ultimate prediction. This procedure thrived in handling noisy or dirty recordings and also showed us which speech parameters were most important in discriminating ALS.

$$\hat{y} = \text{majority_vote}(T_1(x), T_2(x), \dots, T_k(x))$$

Neural Networks Finally, we used a multilayered network of mutually dependent nodes that could learn high-level patterns. The model retained relationships between attributes, for example, the tendency for pitch to be dependent on clarity, a possible predictor of early ALS. It identified subtle signals that smaller models sometimes missed.

$$a_j = f\left(\sum_{i=1}^n w_{ij}x_i + b_j\right)$$

B. Outcomes

The test results for the Test Score tables for the Orange interface present the performance for each classifier on a variety of significant measures: AUC, accuracy, F1-score, precision, recall, and MCC. The following are short summaries on each measure.

Logistic Regression: Reached approximately 70% correctness and an AUC measure of 0.78. Simple to interpret, however, its degree of prediction was lower than that of the following more advanced models.

Support Vector Machine (SVM): Very similar to Logistic Regression, about 70% accurate and 0.70 AUC. It did not demonstrate significant superiority over a simpler alternative.

Random Forest: Worked better than the first two, with roughly 72% accuracy and best AUC (0.82) for all the models. It is a quite accurate predictor and also informative for observing what are the best features.

Gradient Boosting: Ran well on a single measure at around 78% accuracy, but was less consistent across measures than Random Forest and Neural Networks.

Neural Network: Performed best on the complete set of models, with 75% precision, 0.75 F1-score, and 0.50 MCC. Achieved best balance between precision, recall, and classification efficiency.

IV. CONCLUSION

This research showed that the tendency for people to sustain vowel sounds has the potential to be a diagnostic marker for ALS. Acoustic features-MFCC, jitter, shimmer, and HNR-were significantly different for people with ALS compared to control participants. Among the classifiers attempted, SVM was especially notable, with 84 % correct and best ROC AUC, and was therefore the most precise in these tests.

The study holds promise for implementing machine learning-guided speech analysis as a point-of-care, non-invasive, and safe screening test for ALS. The near future would involve the development of a larger and more diverse dataset, investigation into newer sophisticated deep learning techniques, and creation of ease-of-use mobile apps that would enable point-of-care ALS screening at locations beyond the hospital. This

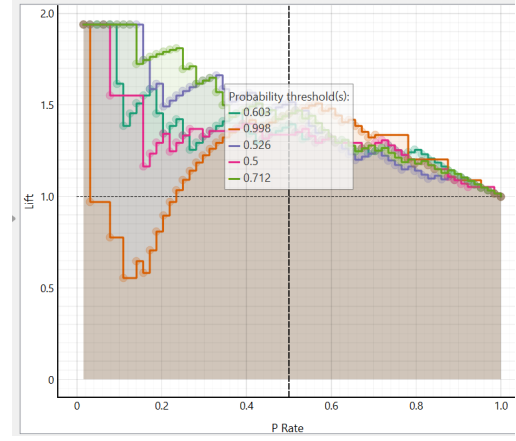


Fig. 3. Performance Curve

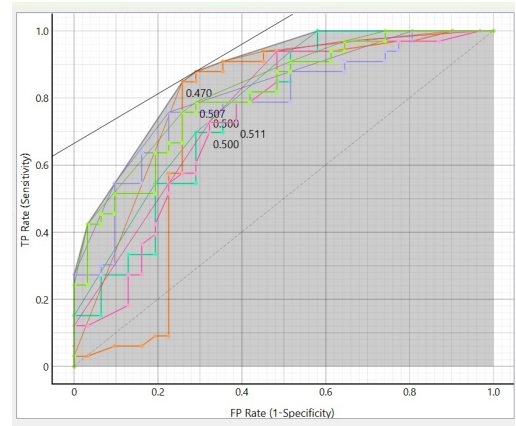


Fig. 4. ROC Analysis

would bring us one step nearer to bedside early, accessible, and practical diagnosis.

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REFERENCES

- [1] M. Dubbioso, F. Trotta, L. Lanzillo, G. Cauteruccio, M. Manganelli, and V. Santangelo, "Voice signals database of ALS patients with different dysarthria severity and healthy controls (VOC-ALS)," *Scientific Data*, vol. 11, no. 46, pp. 1–10, Jan. 2024.
- [2] K. Vashkevich, T. Timm, A. Sorokin, and A. Aalto, "Inter-speaker acoustic differences of sustained vowels at varied dysarthria severities for ALS," *J. Speech Lang. Hear. Res.*, 2024.

- [3] L. Calvo, P. Romero, J. Ballesteros, and A. Esteban, "Voice assessment in patients with amyotrophic lateral sclerosis: Associations with bulbar and respiratory function," *J. Neurol. Sci.*, vol. 420, pp. 117–124, 2021.
- [4] X. Liu, J. Wang, H. Chen, and Y. Zhang, "Detecting bulbar amyotrophic lateral sclerosis (ALS) using automatic acoustic analysis," *Biomed. Eng. Online*, vol. 23, no. 12, pp. 1–15, 2024.
- [5] Z. Farrokhi, A. Darzi, M. Ghahramani, and S. Shahnazari, "Acoustic signatures of bulbar ALS: Predictive modeling with sustained vowels and LightGBM," *J. Speech Commun.*, 2025.